

Highlights

Domain-Validity-Gated Metamorphic Testing of Scientific ML Surrogates

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- A domain-validity rubric screens candidate MRs for SciML surrogate testing.
- MR tolerances are gated by the measuring operator's floor ($O(h)$ for divergence).
- MR cards convert retained candidates into auditable executable test assets.
- A fail-closed claim ledger ties every reported number to a tracked artifact.
- K=6 rosters expose where MR detectors are structurally insensitive.

Domain-Validity-Gated Metamorphic Testing of Scientific ML Surrogates

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Abstract

Context: Scientific machine-learning (SciML) surrogates approximate expensive simulations, but exact outputs for arbitrary inputs are unavailable (the oracle problem). Metamorphic testing addresses this through relations across executions, yet a candidate relation is not automatically valid: its preconditions, output mapping, and the scoring operator’s numerical floor determine whether a violation is meaningful.

Objective: We investigate how candidate metamorphic relations (MRs) can be screened for *domain validity* and operationalized as executable, oracle-free test assets for SciML surrogates, making the candidate-to-asset step explicit about the physical, numerical, and software conditions each relation requires.

Method: We propose (i) a domain-validity rubric admitting a candidate only when its tolerance dominates the operator’s numerical floor and its preconditions hold; (ii) an MR-card executable-asset format recording transformations, mappings, metrics, tolerances, exclusion rules, and relation-level verdicts; and (iii) a case-study protocol on MeshGraphNets cylinder-flow surrogates, with a claim ledger binding every result to an artifact.

Results: On a MeshGraphNets checkpoint, node permutation holds to machine precision, mirror-y is a bounded OOD-stress finding rather than an admitted symmetry, and absolute conservation stays deferred while a

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reference-relative guard passes; these readings hold across the roster, further architectures, and PhysicsNeMo; the same predicate transfers to FNO surrogates. On a second CFD task the gate instead rejects incompressible continuity on physical grounds, reasoning about validity rather than running a checklist. The same gate also fixes what the suite *detects*: used as fault detectors, coverage is generated by the admissible set, which a cross-SUT test confirms.

Conclusion: The same physics-based gate that admits a relation also fixes what it can detect: validity and fault-coverage are two faces of one gate. We state this duality as a falsifiable principle: excluding a relation removes exactly its coverage, which the cross-SUT experiment confirms, and we set out what would refute it. Broader generalization is future work.

Keywords: Metamorphic testing, metamorphic relation identification, oracle problem, scientific machine learning, neural surrogates, MeshGraphNets, domain-validity rubric

1. Introduction

Scientific computing increasingly uses learned surrogates to reduce the cost of repeated numerical simulation. Mesh-based neural simulators such as MeshGraphNets learn to predict physical fields on irregular meshes [1]. These models approximate high-cost solvers but introduce a familiar software testing problem: for many candidate inputs there is no cheap exact expected output against which the program can be checked, the test oracle problem [2].

The common response is to evaluate on held-out trajectories and report rollout error. This is necessary but incomplete: a model may have acceptable average error while failing to preserve relations that should hold under physically meaningful transformations. For a user deploying a SciML surrogate outside a narrow validation set, the question is not only “How accurate is the model?” but also “Under which transformations does the system stop respecting relations it should respect?”

Metamorphic testing (MT) addresses the oracle problem by checking relations among multiple executions [3, 4]. Conservation laws, symmetry relations, nondimensional similarity, and continuity conditions all suggest possible relations among executions [5, 6, 7]. The difficulty is deciding which candidates are valid and executable for a particular SciML program: for

cylinder flow, mirror symmetry depends on geometry and boundary labels; a divergence check depends on a discrete operator, mesh weights, and boundary treatment. Treating such transformations as automatically valid MRs hides the assumptions that determine whether a violation is meaningful.

The remaining problem addressed here is therefore narrower: how candidate MRs are screened for domain validity, turned into executable test assets, and reported through relation-level verdicts that distinguish SUT inconsistency from out-of-relation-domain cases and numerical tolerance effects.

This paper treats MR identification for SciML as a validity-gated testing problem. Physical knowledge, expert reasoning, LLM-assisted candidate lists, and NOETHER-style pattern organization [8] can all suggest candidate relations, but a candidate is not yet an executable MR: it must first state the physical basis, transformation preconditions, boundary-condition compatibility, output mapping, metric, tolerance rationale, exclusion rule, and verdict interpretation.

Two ideas organize this treatment. First, a candidate is *admissible* only when numerically decidable: its verdict tolerance must dominate the intrinsic error floor of the measuring operator, machine precision for an exact relation, the mapping floor for an approximate geometric relation, or the discretization floor for a continuity relation. Second, a relation-level verdict is read in two dimensions: violation magnitude and domain-violation magnitude, so a model inconsistency is not confused with a relation applied outside its domain. We refer to this as a domain-admissibility-gated, relation-indexed approach to SciML out-of-distribution (OOD) validation. The primary case study is MeshGraphNets-family cylinder flow, with the same predicate additionally exercised on a second CFD task (compressible airfoil) and a second PDE family (FNO/PINN on Burgers and heat); full cross-SUT rate estimation remains blocked.

1.1. Research questions

The main research question is:

RQ0. How can candidate metamorphic relations for scientific machine-learning systems be screened for domain validity and converted into executable oracle-free test assets without relying on exact per-sample expected outputs?

We decompose this into four questions:

RQ1 – Validity. How can a domain-validity rubric distinguish physically

meaningful candidate MRs from transformations that are executable but invalid, underspecified, or outside the relation’s domain?

RQ2 – Operationalization. How can retained candidates be represented as MR cards and executable assets with source cases, follow-up transformations, metrics, thresholds, exclusions, and verdict rules?

RQ3 – Verdict. How can relation-level verdicts distinguish pass, fail, skip, out-of-relation-domain, numerical tolerance issue, and inconclusive outcomes?

RQ4 – Case-study evidence. In a MeshGraphNets-family cylinder-flow case study, what evidence does the rubric-gated asset workflow add relative to rollout-accuracy diagnostics, secondary LLM/generic candidate baselines, and external witness evidence?

1.2. Contributions

This paper makes three scoped contributions, each repositioned narrowly with respect to the closest prior identified in Section 2.4. The positive claim is methodological: the paper contributes a measurement-floor admissibility gate, a typed domain-inadmissibility verdict, and a predictable-coverage result from the seeded-fault diagnostic stress test: the same gate that types verdicts also predicts which faults the MR suite can and cannot detect, while the MR-class-to-fault-class diagnostic is reported as stress-test evidence, not a validated localization model.

Measurement-floor admissibility gate. A domain-validity rubric and MR-card executable-asset workflow screen candidate MRs and convert retained ones into auditable executable test assets recording physical basis, transformation preconditions, output mapping, metric, tolerance, and exclusion rules. We ground the tolerance floor in the *intrinsic error floor of the discrete measurement operator* (theoretically $O(h)$ for a P1 divergence operator on a triangular mesh); for the concrete deployed-scale mesh this floor is closed-form, a leading-order predictor matches the measured floor to within 0.5% and a rigorous a-priori bound dominates it (Sec. 5.6), with a general unstructured-mesh bound left to future work. To our knowledge, grounding

an MR tolerance in the measurement operator’s own error floor is new in the SciML setting.

Typed domain-inadmissibility verdict. A relation-level verdict and ledger scheme reads verdicts in two dimensions (relation-violation against domain-violation magnitude). This instantiates, for physics-governed SciML, the constraint-architecture pattern of Duque-Torres et al. [9, 10] and MetaTrimmer [11], adding a *typed classification* of domain-inadmissibility sub-dimensions drawn from PDE preconditions, geometry, boundary conditions, and operator admissibility.

Predictable detector coverage. The seeded-fault diagnostic stress test (the paper’s own MRs used as detectors against an independently re-implemented 10-mutant catalogue) exposes a coverage law: each admissible MR scores one invariant, so the suite catches a fault exactly when it perturbs a measured invariant and is blind to it otherwise. The admissibility gate therefore predicts the suite’s blind spots, confirmed by two falsifiable predictions: detection collapses at the knife-edge where a fault becomes simultaneously permutation- and mirror-symmetric, and removing an inadmissible MR on a second SUT removes exactly its coverage. The class-to-class mapping this produces is consequently SUT-specific, reported as stress-test evidence rather than a localization model.

To keep the contribution boundary easy to follow: the three devices above are the claims; the within-SUT pilots, replication, operator-floor, and fault-detection subsections carry their primary evidence; the cross-family subsections show the predicate transferring across model families; everything else (baselines, witness audits) is context, not claim.

We evaluate these contributions on one task with deliberately wide architectural coverage: a trained MeshGraphNets checkpoint, its K=6 roster (six checkpoints) over three official held-out trajectories, same-domain S4/S5 wider/deeper variants, a non-message-passing PointMLP network, and the NVIDIA PhysicsNeMo production implementation evaluated on official DeepMind cylinder_flow data. Cross-family PINN and FNO executions check that the same admissibility predicate transfers outside mesh surrogates, and rollout-accuracy, exact mirror-y, Minimum-MR-SubSet rerun, and generic-MR / LLM-candidate comparators complete the design.

2. Background and Related Work

2.1. *Mesh-based neural simulation*

Mesh-based neural simulators learn dynamics on graph or mesh representations of physical systems. MeshGraphNets encodes mesh nodes and edges, propagates information by message passing, and predicts future physical fields through autoregressive rollout [1]. The cylinder-flow benchmark exposes the distinction between data-driven accuracy and physical relation preservation. In this paper, MeshGraphNets-family systems are treated as software systems under test: we ask how outputs behave under controlled input transformations and whether necessary relations hold within explicit tolerances.

2.2. *Metamorphic testing and the oracle problem*

Metamorphic testing (MT) addresses programs lacking a per-input oracle [3, 4, 12]: instead of checking each output against an expected value, MT checks a relation among outputs of a source and follow-up inputs. This framing has been applied to scientific software, simulations, and ML systems [13, 5, 7], recently including tabular ML models such as credit scoring [14]. For SciML surrogates the oracle problem is especially visible in OOD settings, where a trusted solver may be too expensive to run for every transformed case. SciML MRs cannot be treated as generic input perturbations: every transformation must respect governing equations, boundary conditions, mesh representation, and numerical tolerance, making MR correctness a central validity question.

2.3. *MR identification for scientific and simulation software*

MRs can be elicited from monotonicity, conservation, scaling, and domain-specific expectations [5, 6, 7, 15]. Predictive MR identification has been pursued with graph-kernel learning over scientific source code [16], treating MR discovery itself as a learning task; large language models have since been used to discover metamorphic-relation variables for scientific software [17]. Data-driven search can also discover candidate transformations in complex physical software [18], and design assumptions can be encoded as MRs for CPS testing under control-plant invariants [19]. The gap is that candidate identification is insufficient for SciML: without a domain-of-validity record a failed test is ambiguous: the relation may have been invalid under the chosen boundary conditions, the transformation may have left the physical regime,

the tolerance may not have been numerically justified, or the SUT may have violated a genuine constraint. This paper makes that distinction explicit.

2.4. *Physics-based MT for learned scientific simulators*

Three contemporary works are the closest prior. **Reichert et al.** [20] applied physics-derived MRs to a trained LSTM hydrologic surrogate, stratifying pass/fail by basin elevation and informally filtering basins where forcing uncertainty dominated the response. We formalise these practices as an explicit admissibility predicate and a two-dimensional verdict, and extend them to mesh-based neural fluid surrogates whose MR validity depends on geometry and discrete operators.

Eniser et al. [21] introduced *relaxations*, numerical tolerances embedded in MR oracles, for stochastic RL policy testing. We extend this calibrated-tolerance principle to deterministic surrogates by grounding the tolerance floor in the intrinsic error floor of the discrete measurement operator (theoretically $O(h)$ for a P1 divergence operator); for the concrete deployed-scale mesh the floor is given in closed form (a leading-order predictor within 0.5% plus a rigorous a-priori bound; Sec. 5.6), with a general unstructured-mesh bound left to future work.

The 2023 violation-attribution cluster, Duque-Torres et al. [9, 10] and MetaTrimmer [11], addressed the bug-vs-inapplicability separation with explicit MR constraints and automated constraint derivation. Our two-dimensional verdict instantiates this pattern for physics-governed SciML, adding a *typed classification of domain-inadmissibility sub-dimensions* drawn from PDE preconditions, geometry, boundary conditions, and operator admissibility. The element none of these three works addresses is the seeded-fault MR-as-detector with by-class localization of Section 5.3.

PINN training itself exhibits gradient-flow pathologies that bias trained models away from the governing residual [22, 23], sharpening the case for relation-level validation that does not assume a converged surrogate.

2.5. *SciML V&V, residuals, UQ, and failure modes*

SciML reliability research has developed complementary tools: physics residuals, uncertainty quantification, conformal prediction, certified error bounds, and failure-mode analysis [24, 25, 26, 23, 27]. A residual or equivariance error becomes part of an executable MR only when paired with a

Table 1: Closest-prior positioning. Each prior work contributes one ingredient of validity-gated MR identification. What this paper adds: it combines them and grounds the tolerance in the operator’s own error floor, types the two-dimensional domain-inadmissibility verdict, and maps MR classes to fault classes (a SUT-specific stress test, not a validated localization model), executed end-to-end on physics-governed SciML surrogates (the prior works are applied on their own domains: hydrologic LSTM, stochastic RL, general MR pipelines).

Approach	Admissibility filter	Floor-grounded tolerance	Typed (2D) verdict	MR→fault-class
Reichert et al.	informal	—	—	—
Eniser et al.	—	relaxation only	—	—
Duque-Torres / MetaTrimmer	constraint-based	—	1D (bug vs. inapplicable)	—
This paper	four-condition rubric	$O(h)$ operator floor	yes	by-class

source case, a follow-up transformation, a tolerance rule, and a verdict interpretation. We use SciML diagnostics as possible relation measurements; MT supplies the multi-execution oracle-free structure.

2.6. Hybrid ML-solver trust regions

Hybrid ML-solver frameworks use residual thresholds or error estimates to decide when a learned component should be trusted [28, 29]. Such systems operationalize runtime trust regions, whereas the present study acts offline: physically derived transformations are applied before deployment to estimate where relation violations occur.

The distinction from residual- and uncertainty-based trust estimation is deliberate. UQ, conformal prediction, residual-threshold trust regions, and physical-consistency diagnostics that score a physically derived residual on observed outputs [30] locate unreliable behavior passively from observed inputs. The present method acts in relation space: it applies a controlled transformation and reports which necessary relation breaks under it, indexed by that transformation. The two-dimensional verdict separates a model-level violation from an out-of-domain application, a distinction a scalar accuracy or residual magnitude does not by itself provide. Section 5.3 gives the concrete

instance: a surrogate accurate in-distribution (median one-step relative L2 0.0216) still violates mirror-y equivariance by roughly an order of magnitude, so accuracy does not bound the relation violation; the gate-admitted MR surfaces a property failure that an in-distribution accuracy or UQ monitor, reading only output magnitude, does not, which is the complementarity the case study reports rather than a superiority claim.

2.7. *What is new and what is not new*

Metamorphic testing, MR identification, scientific-software MT, residual diagnostics, uncertainty quantification, LLM candidate generation, and NOETHER-style candidate organization are established or emerging sources of testing ideas. The narrower claim is that SciML MR identification should be treated as a domain-validity problem: a candidate relation becomes useful only after its physical basis, transformation preconditions, output mapping, tolerance, exclusion rule, executable artifact, and relation-level verdict are recorded. Physics-based MRs for learned field predictors are an active area; the verified record sharpens the novelty statement toward validity-gated execution rather than first-use positioning.

What is new is the evidence-gated conversion from candidate relation to executable SciML MR asset, to our knowledge the first end-to-end validity-gated metamorphic-testing pipeline for physics-governed SciML in which every stage, from candidate screening through verdict typing to fault-class diagnosis, is operationalized and executed rather than proposed. Table 1 positions this contribution: each prior supplies one ingredient, and none combines them. The structural novelty is two organizing devices: an admissibility gate tying a relation’s tolerance to the numerical error floor of its own measurement, and a two-dimensional verdict separating a model violation from an out-of-domain application; the same predicate executes unchanged across MeshGraphNet, PointMLP, PhysicsNeMo, PINN, and FNO subjects. The admissibility gate is fully operational in the case study; the verdict’s domain-violation axis is operationalized per relation as a continuous score from committed precondition measurements (Section 3.6); calibrating these scores across MR classes is still future work. These two devices in turn expose the paper’s central claim, a *validity-coverage duality*: the same physics-based gate that decides whether a relation yields a meaningful verdict also fixes which faults it can detect, so it generates the suite’s coverage and blind spots (Section 5.7, claim C37). We state it as a falsifiable principle:

qualitative, evidenced on two CFD tasks, stated to be refuted rather than admired.

3. Method

3.1. Overview

The proposed method is a five-stage workflow: (1) identify candidate relation sources; (2) organize candidates using declared source categories; (3) screen with a domain-validity rubric; (4) convert retained relations into executable MR assets; (5) execute assets and record relation-level verdicts. Candidate generation is deliberately separated from validity judgment: NOETHER-style patterns may help organize candidates but do not certify that a relation is physically valid for a particular SUT. Figure 1 summarizes the workflow.

3.2. Candidate relation sources

Candidate MRs are read off the algebraic properties of the governing and discrete operators, each located in one of three levels (physical-model, computational-model, code-model):

- **equivariance:** reflections and rigid motions give physical-model symmetry MRs under compatible geometry, and node or face permutation gives code-model representation contracts.
- **conservation:** discrete-divergence and boundary-flux relations, decidable only when the operator floor permits (Sec. 5.6).
- **homogeneity/scaling:** Reynolds and Strouhal relations, within regime.
- **composition:** rollout determinism and prefix-consistency, which are implementation checks rather than physical laws.
- **cross-implementation comparison:** applicable only when units, meshes, rollout horizons, and checkpoints are comparable.

The same property can hold at the physical-model level yet fail at the computational or code level: an asymmetric mesh breaks reflection-equivariance, and an uncalibratable operator floor blocks conservation decidability. The admissibility gate therefore locates where each property survives, and the typed verdict records it.

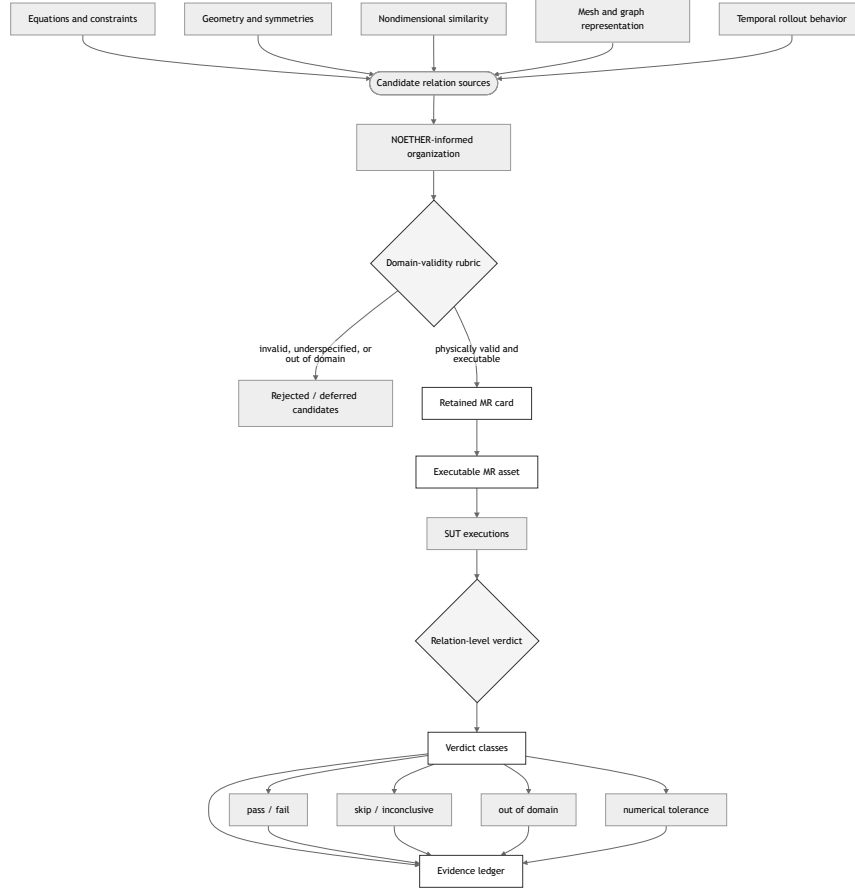


Figure 1: Validity-gated workflow for converting physically motivated candidate relations into executable, oracle-free MR assets and relation-level verdicts. The rubric and the verdict are explicit gates (diamonds); rejected candidates and out-of-relation-domain cases are recorded as evidence rather than discarded silently.

Table 2: Domain-validity rubric for candidate MRs.

Criterion	Question asked before retention
Physical basis	What equation, boundary condition, nondimensional law, representation property, numerical assumption, or implementation contract justifies the relation?
Transformation preconditions	What exactly changes from source to follow-up, and what must be preserved?
Boundary-condition compatibility	Does the transformation preserve the intended physical meaning of the domain and boundaries?
Output mapping	Can the expected output relation be measured from available SUT outputs?
Metric and tolerance	What metric is used, and where does the tolerance come from?
Failure diagnosability	What can a violation plausibly mean, and what can it not mean?

3.3. Domain-validity rubric

A candidate relation is an *admissible MR* when four conditions hold together: (i) it has a physical or software basis; (ii) its transformation preconditions are satisfied; (iii) boundary conditions and output mapping remain compatible after the transformation; and (iv) it is numerically decidable, the verdict tolerance dominates the intrinsic error floor of the measuring operator. None of the four is optional: each is a gate that can reject or defer a candidate. The six rubric criteria in Table 2 are the auditable form of these four conditions.

The rubric yields one of four decisions: retained as executable MR; retained as OOD stress relation; rejected; or deferred pending missing evidence. The defer and OOD-stress decisions are the two soft-fail paths. A candidate is deferred when condition (iv) cannot yet be established, for example an absolute discrete-divergence relation whose reference field already carries non-negligible divergence. It is downgraded to OOD-stress when conditions (ii)–(iii) hold only approximately.

3.4. MR card and executable asset format

A retained MR is represented as an MR card recording: identifier, source category, physical or software basis, source-case schema, follow-up transformation, preconditions, boundary-condition compatibility, output mapping, metric, tolerance and provenance, expected verdict classes, exclusion rules, artifact schema, and fault interpretation. Initial card skeletons for the cylinder-flow study are listed with their verdicts in Tables A1 and 5.

Executable assets are generated from MR cards; each includes transformation code, runner configuration, metric computation, verdict logic, and artifact recording. Figure 2 shows the resulting data flow.

3.5. Worked examples: from candidate relation to executable MR card

These are design-time screening decisions, not empirical findings, and do not report whether any SUT passes or fails. Table 3 summarizes three such decisions, stressing different parts of the method: representation semantics, physical symmetry under boundary conditions, and a constraint measurement whose validity depends on a discrete operator and tolerance provenance.

MR-1: node permutation equivariance.. A permutation π is applied to node order and corresponding edge indices without changing coordinates, fields, or boundary labels. Equivariance requires that the output for the permuted case equals $\pi(Y)$ up to numerical tolerance; the output mapping applies π^{-1} before comparison. This is a representation-level contract: tolerance is derived from deterministic repeat runs or machine precision; a failure points to graph adapter, batching, or message-passing implementation issues, not a Navier–Stokes violation.

MR-2: mirror-y equivariance.. Coordinates (x, y) map to $(x, -y)$; the transverse velocity component changes sign; boundary labels transform with the mesh. Retained only when cylinder, inlet, outlet, and wall labels remain semantically compatible after reflection; rejected if the transformation swaps non-equivalent boundaries or leaves vector components unmapped. A violation may indicate a SUT symmetry inconsistency or that the MR was applied outside its domain; the relation-level verdict must preserve that distinction.

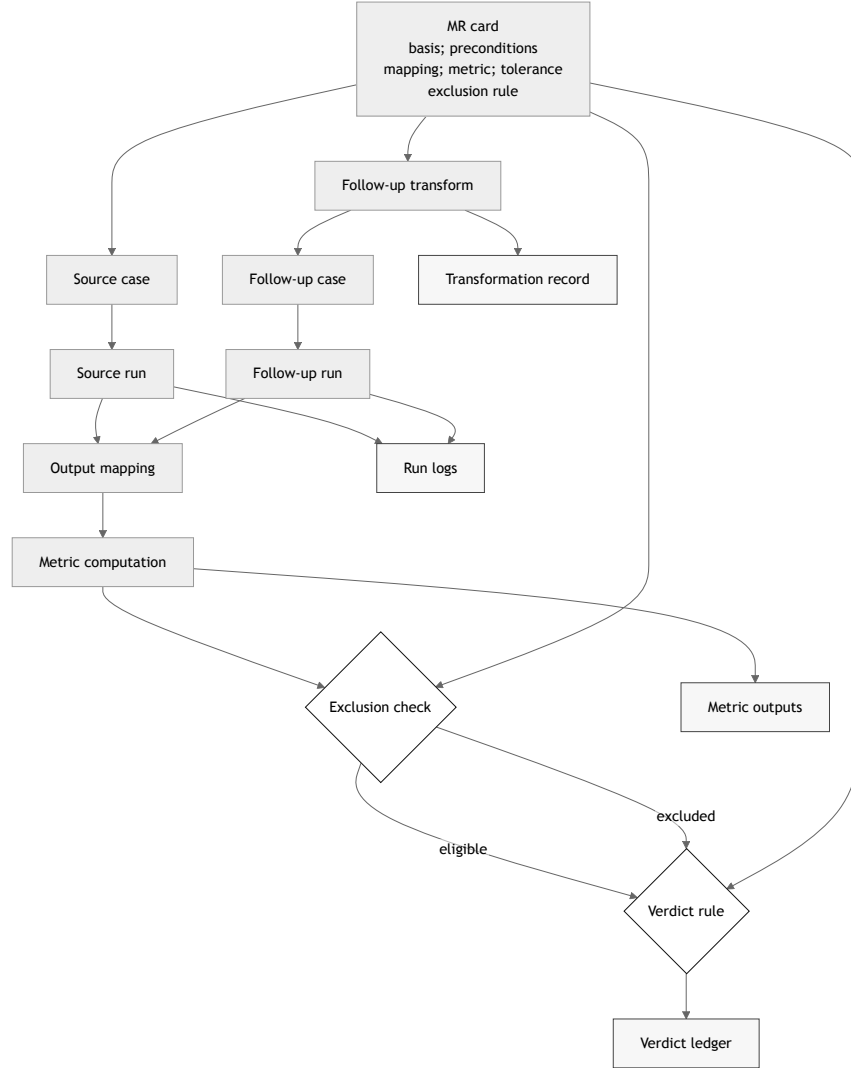


Figure 2: Executable MR asset and verdict data flow. The MR card supplies the basis, preconditions, mapping, metric, tolerance, and exclusion rule; the source and follow-up runs feed the output mapping and metric; the exclusion check and verdict rule emit a relation-level verdict to the ledger together with metric outputs, run logs, and the transformation record as artifacts.

Table 3: Worked screening decisions for three candidate MRs.

Candidate	Main validity issue	Screening decision	Reason
Node permutation equivariance	Graph relabeling must preserve the same physical mesh and fields	Retain as executable representation MR	The transformation changes representation order but not the physical case, provided the adapter and inverse output mapping are deterministic.
Mirror-y equivariance	Geometry, boundary labels, and vector components must transform consistently	Retain only under boundary-compatible conditions	The relation is physically meaningful for symmetric cylinder-flow cases, but invalid if the transformation changes inlet, outlet, wall, or obstacle semantics.
Discrete divergence boundedness	The metric depends on the discrete divergence operator, mesh weights, boundary treatment, and tolerance	Defer or retain as qualified continuity-constraint MR	The physical basis is strong for incompressible flow, but the executable verdict is meaningful only when the numerical operator and threshold are justified.

MR-3: discrete divergence boundedness.. Predicted velocity fields should remain within a bounded discrete divergence tolerance for eligible interior cells. Retained only after the discrete operator, mesh weights, boundary treatment, and tolerance provenance are specified; the exclusion rule skips cases where boundary treatment dominates the metric or where the flow is outside the incompressibility assumption. A failure is a continuity-constraint violation only after operator and tolerance evidence rule out a numerical artifact.

These examples show how the rubric changes the status of candidate relations: node permutation becomes a retained representation MR, mirror-y a conditional physical-symmetry MR, and divergence a qualified continuity-constraint MR whose verdict depends on numerical instrumentation.

3.6. Relation-level verdicts

An MR execution can produce: **pass** (relation holds within tolerance); **fail** (violated within valid domain); **skip** (precondition missing); **out-of-**

relation-domain (transformation produced a case outside the validity domain); **numerical-tolerance issue** (dominated by threshold/resolution uncertainty); or **inconclusive** (insufficient artifact).

These verdicts form a two-dimensional space. One axis is relation-violation magnitude (V/floor ratio); the other is domain-violation magnitude (how far the transformed case lies outside the validity domain). Low domain-violation with high relation-violation is the only region that may be read as SUT inconsistency; high domain-violation is out-of-relation-domain; a violation within the error floor is a numerical-tolerance issue. This decomposition keeps a model-level violation from being confused with a relation applied outside its domain.

In the present study the relation-violation axis is quantitative (mirror-y reports V/floor), and for the mirror-y relation the domain-violation axis is now operationalized as a continuous geometric score $D = m/(m+1) \in [0, 1)$, where m is the worst reflected-node placement error in median-edge-length units (committed; Table A1): the synthetic symmetric mesh scores $D \approx 0$ and the real asymmetric eval mesh scores $D = 0.51$. The same construction covers every executed MR class from committed measurements: permutation-class and PINN closed-form mirror relations are exact-by-construction ($D = 0$), conservation relations score the committed open-boundary flux imbalance (MGN $D = 0.036$; Burgers $D = 0.042$; heat $D = 0$), and one roster-level entry is marked not operationalizable from committed data. The m measures differ in units across classes, so D is a per-relation normalized coordinate, not a calibrated score and not a cross-relation calibrated metric: D values cannot be averaged or ranked across MR classes and should be read per relation, not as a calibrated boundary measurement. The concrete path to cross-relation calibration is to divide each class’s precondition violation by that class’s own admissibility boundary, mapping every gate threshold to a common $D_{\text{cal}} = 1$ (the device that makes the relation-violation axis comparable through V/floor); specifying and validating that normalization across all MR classes and SUTs is left to future work. Figure 3 plots the four cylinder-flow pilots; the synthetic-symmetric-mesh probe sits in the SUT-inconsistency region, which makes the mirror-y finding a model-level result rather than an out-of-domain artefact.

Retained MRs are organized using the three-level metamorphic-relation classification for scientific-computing programs of Yang et al. [31] (physical-model, computational-model, and code-model relations), used as a predeclared interpretation protocol mapping each MR class to its candidate fault

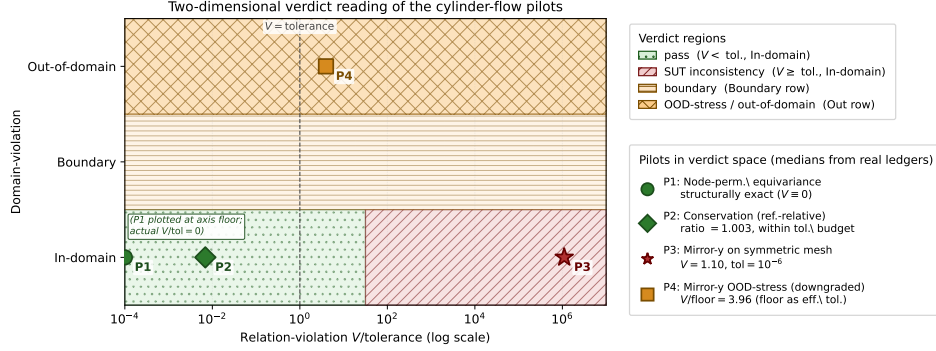


Figure 3: Two-dimensional verdict reading of the cylinder-flow pilots. Horizontal axis: relation-violation $V/\text{tolerance}$ on a log scale, where tolerance is the per-MR verdict threshold (machine-precision floor for an exact relation; the reference-relative regression threshold for conservation; the mapping-error floor when an approximate relation has been downgraded to an OOD-stress probe, as for P4). Vertical axis: domain-violation as a three-level categorical bin (In-domain, Boundary, Out-of-domain); the bins are backed by the per-relation continuous scores of Sec. 3.6, which are not mutually calibrated across relations. The dashed line at $V/\text{tolerance} = 1$ is the pass/fail border for an admissible relation. Coordinates (medians from the metric ledgers; full numbers in Sec. 5.3 and Table 5): P1 node-permutation equivariance $V = 0$ exactly (structurally exact, plotted at the log-axis floor); P2 conservation (reference-relative) ratio = 1.003, so $V/\text{tolerance} \approx 0.01$, two orders of magnitude above P1 yet still well inside pass; P3 mirror-y on the synthetic symmetric mesh, where the rubric admits the exact relation, $V = 1.10$ against tolerance 10^{-6} , $V/\text{tolerance} \approx 1.1 \times 10^6$: the SUT-inconsistency region, which is what removes the out-of-relation-domain objection to the mirror-y finding; P4 mirror-y OOD-stress (downgraded), $V/\text{floor} = 3.96$ with the mapping-error floor as effective tolerance, sitting in the Out-of-domain row. Absolute conservation is not plotted: its admissibility predicate (Sec. 3.3) fails because the absolute tolerance does not dominate the P1 operator floor calibrated in Sec. 5.6, so the rubric never admits it into the verdict space.

layer. This is a localization protocol, not a validated localization model; Section 5.3 reports a first bounded test against an injected-fault catalogue.

4. Empirical Design

4.1. Subject systems

The study evaluates the workflow on three groups of subject systems, all with recorded checkpoints, datasets, seeds, and environments in the experiment ledger. *Cylinder-flow surrogates (primary)*: a trained MeshGraphNets surrogate and its K=6 checkpoint roster, a same-domain S4/S5 wider/deeper variant package, a newly trained non-message-passing PointMLP coordinate network, and the NVIDIA PhysicsNeMo **MeshGraphNet** production implementation evaluated on official DeepMind cylinder_flow TFRecords. *Cross-family replication subjects (supporting)*: a K=15 PINN roster and six trained FNO-2D checkpoints over 2D Burgers and heat data. *External witness runtime (secondary)*: the read-only Minimum-MR-SubSet repository’s held-out MGN and trained-PINN witnesses, rerun locally for provenance. *Cross-program breadth subjects (generalization check)*: seven program types from the read-only Minimum-MR-SubSet sibling (neural surrogates, classical numerical solvers heat/wave/point-kinetics/Burgers, and a production Monte-Carlo code OpenMC), reused read-only as committed kill matrices, with the four classical solvers and the OpenMC *k*-eigenvalue subject additionally executed end-to-end through this paper’s gate and typed verdict on CPU (Section 5.7); these are bounded breadth checks of the coverage-geometry reading, not primary subjects. Because the primary subjects share one task and dataset, the cylinder-flow results are framed as same-task multi-architecture evidence, not evidence for all neural fluid surrogates.

4.2. Planned MR classes

The initial MR set covers representation, geometric/symmetry, continuity/constraint, nondimensional similarity, numerical stability, and temporal/implementation MRs. Time reversal is excluded for viscous cylinder flow; divergence is treated as a continuity constraint, not a Noether-derived conservation law.

4.3. Baselines and comparators

The study uses four secondary comparator families: (1) **LLM-simulated expert MR design**, three expert-LLMs propose candidates independently,

Table 4: Research questions, efficacy parameters, baselines, and artifacts.

RQ	Primary parameters	Comparators	Artifacts
RQ1	candidate retention rate; rejection reasons; inter-rater agreement	expert MR design; LLM candidates	rubric sheets; candidate ledger; adjudication log
RQ2	executable MR rate; transformation success; asset completeness	expert MR cards; generic scope contrast	MR cards; transformation code; runner configs
RQ3	verdict distribution; out-of-domain rate; interpretation yield	residual/UQ diagnostics where available	verdict ledger; metric outputs; exclusion logs
RQ4	violation rate; violation magnitude; complementarity with rollout error; fault-detection rate if seeded faults exist	LLM-simulated expert, LLM, and generic candidates as secondary contrasts; rollout accuracy as diagnostic evidence; external-scope audit as provenance	run logs; plots; confidence intervals; fault ledger

scored by a three-rater majority-vote admissibility panel; (2) **Generic MR-generation scope contrast**; (3) **LLM-assisted candidate generation**; and (4) **Rollout-accuracy diagnostic**. All four are executed (Sections 5.3, 5.8). Residual/UQ metrics are diagnostic comparators, not MRs unless paired with a source/follow-up transformation and explicit verdict rule.

4.4. RQ-to-evidence map

Table 4 maps each research question to the planned evidence. This table is a pre-results design statement rather than a report of findings.

4.5. Efficacy parameters and statistical plan

Primary efficacy parameters are candidate retention rate, executable MR rate, MR violation rate, violation magnitude, fault-detection rate, boundary characterization, and interpretation yield. Secondary parameters include MR construction cost, inter-rater agreement, flakiness rate, localization agreement with seeded-fault layers, and complementarity with rollout accuracy.

The study reports inferential statistics wherever the sampling structure supports them and labels descriptive counts where it does not. Binary detector outcomes over the unified fault catalogue carry Wilson 95% intervals per detector; continuous violation magnitudes use exact Wilcoxon signed-rank tests with a rank-based effect size (Cliff’s δ), following the Empirical Standards; roster medians carry paired bootstrap intervals; and the operator-floor sweep reports a log-log slope with a 95% confidence interval (Section 5.4). Where repeated cells are not independent samples (consecutive frames of one trajectory, or checkpoint-by-trajectory grids within one family), the counts

are reported as descriptive cell summaries and labeled as such. Small-sample results foreground effect sizes and intervals over significance claims.

4.6. *Ethics, integrity, and reproducibility*

The study does not involve human subjects or personal data. All SUT versions, datasets, checkpoints, scripts, and run logs are recorded when licensing permits; failed, skipped, and inconclusive cases remain in the experiment ledger. LLM use is restricted to candidate generation and evidence organization, not validity judgment. No candidate MR is described as valid without a passing rubric decision and MR card; no OOD violation is described as a program fault without supporting verdict evidence. Third-party artifacts are used according to their licenses; any non-redistributable artifact is described with access instructions.

5. Results

The results are organized in three tiers. The *primary* evidence is the cylinder-flow case study: the within-SUT pilots, their K=6 multi-checkpoint and multi-trajectory replication, the operator-floor calibration, the seeded-fault stress test, and the same-task multi-architecture executions including the PhysicsNeMo production-framework workflow. The *supporting* evidence is the cross-family transfer of the rubric to PINN and FNO subjects. The *secondary* evidence is the baseline comparators and external witness audits, which contextualize the primary results without claiming method superiority.

5.1. *Claim-to-evidence map*

The compact claim-to-evidence map binds every claim to a tracked artifact using reviewer-facing claim names; the full table is given in Appendix Appendix A (Table A1), and the authoritative runtime mapping remains in `claim-ledger.yml`.

5.2. *MR-card-to-verdict map*

Table 5: MR-card-to-verdict map.

MR card	Rubric decision	Runtime verdict	Interpretation boundary
Node permutation equivariance	Retained as representation MR	pass sanity; relative L2 = 0.0	Supports one executable representation path; does not establish model reliability.
Mirror-y equivariance (asymmetric eval mesh)	Exact relation out-of-relation-domain; downgraded to approximate OOD-stress	S0 pilot: 10/10 fail; primary upgrade: 180/180 fail across K=6 x 3 trajectories x 10	Bounded within-family OOD-stress violation across held-out trajectories; not by itself exact symmetry, cross-SUT, or geometry-independent evidence.
Mirror-y equivariance (synthetic symmetric mesh)	Exact relation admissible (bijection verified, offset $< 10^{-12}$, type-match 1.0)	S0 pilot: fail, rel L2 1.10; primary upgrade: 18/18 fail across K=6 x 3 input seeds	Exact-symmetry failure where the relation is admissible; synthetic no-obstacle OOD meshes, not accuracy or cross-SUT evidence.
Discrete divergence / conservation	Absolute mass-conservation MR deferred; reference-relative diagnostic retained	S0 pilot inconclusive: reference-relative non-regression guard; primary upgrade: 162/162 pass across K=6 x 3 trajectories x 9	Reference-relative diagnostic only; the absolute conservation relation remains deferred.

MR card	Rubric decision	Runtime verdict	Interpretation boundary
MGN S4/S5 variant workflow	Node permutation admitted; mirror OOD/conservation diagnostic/exact-symmetry decisions recorded	2/2 node-permutation passes; 60/60 mirror OOD failures; 54/54 conservation-diagnostic passes; 6/6 exact-symmetry failures	Same-domain MGN variant evidence, not an external SUT family.
PointMLP cylinder workflow	Node permutation admitted; mirror OOD/conservation diagnostic/exact-symmetry decisions recorded	9/9 node-permutation passes; 10/10 mirror OOD failures; 9/9 conservation-diagnostic passes; 3/3 exact-symmetry failures	Different non-MGN cylinder SUT; not PhysicsNeMo/EchoWave or production CFD evidence.
PhysicsNeMo MGN Object-A smoke workflow	Node permutation admitted; mirror OOD/conservation diagnostic decisions recorded	Node permutation passes on the smoke subset (relative L2 0.0); rollout, mirror OOD, and conservation diagnostic ledgers recorded	Production-framework artifact-chain smoke evidence only; no full-scale production pass/fail rate and not external-aerodynamics evidence.

MR card	Rubric decision	Runtime verdict	Interpretation boundary
FNO periodic translation and conservation	Translation admitted; periodic discrete-conservation MR admitted-with-reference-floor; Dirichlet translation rejected	FNO primary workflow upgrade: 24/24 translation passes, 24/24 conservation failures, and 6/6 rejected Dirichlet exact-MR executions	Full rubric-to-verdict FNO evidence with raw source/follow-up outputs and per-case ledgers; outside cylinder-flow and broad neural-operator claims.

5.3. Within-SUT pilot evidence

All pilots run on one real trained MeshGraphNets cylinder-flow surrogate, using the read-only Minimum-MR-SubSet checkpoint recorded in the experiment ledger. They exercise three different rubric outcomes; raw outputs, manifests, and metric ledgers are committed (Table A1).

Representation MR. Node-permutation equivariance holds to machine precision (relative L2 = 0.0 at tolerance 1e-6). This is a structural property of message-passing and is reported as a pipeline-implementation sanity check rather than model capability or accuracy evidence.

Geometric MR. The rubric classifies exact mirror-y equivariance as out-of-relation-domain for this mesh (the reflection is non-bijective, the worst reflected-node mismatch is about one edge length, and the cylinder is off-centre by 7.2 mm) and downgrades it to an approximate nearest-neighbour OOD-stress probe scored against a same-space mapping-error floor. The probe failed on 10 of 10 recorded eval frames (median relative L2 0.737, median V/floor 3.96, floor range 3.02–5.55x), with no frame skipped or inconclusive. This is a bounded within-SUT frame-level OOD-stress result: one SUT, one checkpoint, one MR, one eval trajectory. The frames are consecutive, not independent samples, so the exact binomial 95% interval for 10/10 is wide ([0.69, 1.00]); and because the mapping-error floor derives from the same geometric mismatch that triggered the downgrade, V/floor alone cannot separate a model-level violation from an amplified geometric artifact on this mesh; the symmetric-mesh run below resolves that ambiguity.

Continuity MR. A P1 discrete-divergence operator yields a non-negligible divergence even for the ground-truth field on this coarse mesh, so

an absolute divergence-free tolerance is not calibratable and the absolute mass-conservation relation stays deferred. As a reference-relative diagnostic, the surrogate’s predicted next-state divergence stays within about 0.4–0.8% of the reference on the recorded eval frames; the interior-only ratio confirms this is not only a boundary-imposition artifact. This asserts no absolute conservation. Two caveats bound the diagnostic: the reference divergence is not yet decomposed into discrete-operator error, solver projection artifact, or genuinely non-solenoidal training data, so if the reference field is itself materially non-solenoidal the reference-relative ratio is a non-regression guard rather than a conservation measurement; and the diagnostic uses a 50% regression threshold (ratio > 1.5) on two eval frames only, so “passes” means “does not regress conservation by more than 50% on those frames,” not “conserves mass.”

Rollout-accuracy diagnostic. On the same eval trajectory, the surrogate’s one-step next-state prediction error ($v_{\text{pred}} = v_t +$ denormalized predicted delta, the trainer’s own convention) has median relative L2 0.0216 (mean 0.044; min 0.0116, max 0.0788) over the nine recorded transitions. The per-transition error is bimodal, so we quote both statistics. The mirror-y OOD-stress violation (median 0.737) is about 34 times the median accuracy and 17x the mean. The quantities are dimensionless relative L2 values of different objects, so this is an order-of-magnitude gap rather than a precise factor; it is one-step, not a free-running rollout-stability result.

Exact mirror-y on a symmetric mesh. We built a provably symmetric synthetic structured channel mesh whose reflection is a verified bijection (node-type match 1.0, reflection offset $< 10^{-12}$, edge set invariant), so the predicate admits exact mirror-y. On one constructed input the surrogate fails this exact equivariance test (relative L2 1.10). The normalizer control changes the violation only from 1.1032 to 1.1014, so the violation is dominated by learned message-passing weights. The boundary is synthetic, no-obstacle OOD geometry: the result confirms missing exact equivariance but is not a calibrated in-distribution magnitude.

Seeded-fault detection. We re-implemented, from the read-only Minimum-MR-SubSet witness taxonomy, an injected-fault catalogue of 10 pipeline faults across five fault classes (boundary-condition, mesh-adjacency, normalization-scale, temporal-rollout, physical-channel), and used the paper’s own MRs as detectors on the model’s predicted update. The continuity MR detected the two boundary-condition faults and the gross normalization fault (divergence ratio 3.8–10.6 vs the 1.5 threshold); the symmetry MR

detected a physical-channel and a mesh-adjacency fault (violation rising 69–142% above baseline); node-permutation equivariance detected none, because these faults preserve node-relabeling invariance. 5 of 10 mutants were detected by at least one MR, and the detections separate by MR class (continuity to boundary/scale faults, symmetry to physical-channel/adjacency faults), a first bounded test of the interpretation protocol of Section 3.6, suggestive rather than a validation. Three boundaries: the detected faults are gross corruptions that any divergence- or symmetry-sensitive detector would catch; the edge-drop fault is a near-miss (32% mirror-y change vs the 50% threshold) and the boundary faults are invisible to mirror-y by construction; and the remaining undetected faults shift the absolute output without crossing the scored-quantity thresholds, delimiting where these MRs are structurally insensitive. Against a rollout-accuracy monitor on the same faults the two are complementary: the edge-permutation and channel-swap faults the symmetry MR catches leave the rollout error within $1.3\times$ of its in-distribution baseline (0.0216, under a $2\times$ regression threshold), so the MRs surface relation violations an accuracy monitor leaves within band, while the gross boundary/normalization faults trip both and no fault is caught by accuracy alone (claim C38). It is one SUT, one checkpoint, one injected-fault catalogue.

Component	Evidence	Headline result	Boundary
MGN roster	K=6 checkpoints	Permutation exact; mirror-y median CI [0.743, 0.804]; rollout CI [0.0217, 0.0224]	One architecture family / dataset
Operator floor	P1 symmetric-mesh sweep	O(h), slope 0.984, CI [0.975, 0.992]; closed-form floor for this mesh (predictor within 0.5% + a-priori bound)	Calibrates decidability; concrete-mesh closed-form floor, general unstructured bound is future work
Fault catalogue	cata- 60-entry unified fault catalogue	Precision/recall: permutation 1.00/1.00; conservation 1.00/0.81; mirror-y 0.94/0.55	node-Catalogue rate, not real-world defect rate
PINN roster	K=15, two PDE families	MR-C: Burgers 15/15, heat 15/15 (Wilson CI [0.80, 1.00]); MR-B: Burgers 13/15 (CI [0.62, 0.96]), heat 7/15 (CI [0.25, 0.70])	Not a PINN vs-MGN benchmark
FNO primary workflow grade	K=6, two up-families	24/24 translation passes; 24/24 conservation failures under a periodic discrete-conservation MR; Dirichlet translation rejected 6/6	not only admissibility evidence; outside cylinder-flow and broad neural-operator claims

5.4. Multi-checkpoint, multi-trajectory, and multi-architecture replication

The single-checkpoint evidence is replayed on a K=6 MeshGraphNets roster and three official DeepMind held-out test trajectories acquired through the Minimum-MR-SubSet loader. S0 reuses the pilot checkpoint, so this is a replication/variant check rather than six independent SUTs. Node permutation is exact on all checkpoints. The primary empirical scope upgrade gives a K=6 x 3 trajectories x 10 mirror-y OOD-stress grid with 180/180 failures, a K=6 x 3 trajectories x 9 conservation-transition grid with 162/162 reference-relative passes while absolute conservation remains deferred, and a K=6 x 3

exact-symmetric-mesh input grid with 18/18 failures. The upgrade removes the single-source-trajectory denominator while remaining within one architecture family and dataset, so per-cell counts are descriptive summaries, not independent-trial inference.

Same-task multi-architecture replication. Three further cylinder-flow executions test whether the verdict pattern is an artifact of one implementation. A same-domain S4/S5 wider/deeper MeshGraphNet variant package reproduces the pattern (2/2 node-permutation passes, 60/60 mirror OOD failures, 54/54 conservation-diagnostic passes, 6/6 exact-symmetry failures). A newly trained PointMLP coordinate network (a different architecture class with no message passing) reproduces it as well (9/9 node-permutation passes, 10/10 mirror OOD failures, 9/9 conservation-diagnostic passes, 3/3 exact-symmetry failures, median rollout relative L2 0.0298). Third, the NVIDIA PhysicsNeMo MeshGraphNet production implementation, trained and evaluated on official DeepMind cylinder_flow TFRecords, reproduces the node-permutation pass and the mirror-y OOD-stress failure within a production framework rather than a research codebase. The same admissibility predicate, applied unchanged, produced the same typed decisions on all three; this replicates the verdict pattern across architectures on one task and dataset, and asserts no cross-dataset rate.

5.5. Second task: compressible airfoil flow

The strongest test of a domain-validity gate is whether it produces the correct *different* verdict when the physics changes. We apply the identical four-condition predicate to a genuinely second CFD task on a second official dataset: the DeepMind **airfoil** benchmark (SU2-simulated compressible transonic flow, 5,233-node meshes with density, pressure, and velocity), distinct from incompressible cylinder flow in both physics and data. The same official PhysicsNeMo MeshGraphNet (2.33M-parameter official configuration: hidden width 128, 15 processor steps) is trained for 15 epochs on 25 official airfoil train trajectories across a K=6 checkpoint roster and evaluated per-trajectory on 40 official test trajectories, yielding 240 (checkpoint, trajectory) pairs. The predicate yields a *different typed inadmissibility structure*, and the difference is the result: **node-permutation equivariance is admitted and exact** (240/240, relative L2 0.0, training-independent representation contract); **incompressible divergence-free continuity is rejected at the physical-basis gate** (condition i) because the density varies by a median factor of 2.13x across the field, so $\nabla \cdot u = 0$ is physically false here, the same

relation that on cylinder flow passes conditions (i)–(iii) and is only *deferred* at condition (iv), so the gate assigns a categorically different verdict *type* for the physically correct reason on each task; **compressible unsteady mass conservation** $\partial_t \rho + \nabla \cdot (\rho u) = 0$ is the physically-correct replacement, its absolute discrete form deferred (uncalibratable operator floor) and a reference-relative diagnostic recorded (median residual ratio 1.00 across the K=6 roster); and **mirror-y chord symmetry is rejected at the boundary-precondition gate** (condition iii) because the SU2 trajectories carry a non-zero angle of attack. The one-step rollout error is high at this training budget (median relative L2 1.00, comparable to the signal magnitude) and is reported only as a training-state diagnostic; notably, none of the four typed verdicts depends on rollout quality: the admission rests on a representation contract and the rejections/deferral on physical and numerical reasoning, which is why the typed structure transfers and discriminates across tasks. This is a primary-scale second-task roster (official MeshGraphNet architecture, 240 checkpoint-by-trajectory cells), not an official-checkpoint or SOTA-accuracy airfoil result.

5.6. Operator-floor resolution sweep (calibration of numerical decidability)

The P1 discrete-divergence sweep on symmetric meshes, over nine resolutions, gives slope 0.984 with 95% CI [0.975, 0.992] and $R^2 = 0.9999$, matching $O(h)$. Beyond the slope, the *absolute* floor is closed-form for the concrete deployed-scale mesh: since the operator is exact on affine fields and the reference field is divergence-free, the measured floor equals the geometry-weighted second-order Taylor remainder exactly. A leading-order predictor from the analytic Hessian matches the measured floor to within 0.5% (1.337 vs 1.343), and a rigorous a-priori upper bound from the Hessian’s global spectral norm dominates it at every resolution. The gate thus rests on a derived floor, not an empirical estimate, for this mesh (a general unstructured-mesh bound is future work). This is why absolute conservation remains deferred while the reference-relative non-regression guard is reported. A classical flux-form finite-volume operator instead admits a fully executable conservation verdict (claim C34, supporting artifact).

5.7. Fault-detection robustness across the multi-checkpoint roster

To check whether the seeded-fault detection result of Section 5.3 reproduces beyond a single checkpoint and to delimit where its detectors are structurally insensitive, the 10-mutant catalogue is replayed against the K=6

multi-checkpoint roster across five input-permutation seeds (30 trials per mutant per detector), and two predeclared severity dimensions are swept: NS_double_scale $s \in \{1.1, 1.25, 1.5, 2, 4\}$ and PC_zero_vy partial fraction $p \in \{0.25, 0.5, 0.75, 0.85, 0.9, 0.95, 0.99, 1.0\}$ (the canonical mutant is $p = 1.0$). Detection is decided by the same predeclared thresholds as Section 5.3 (node-permutation tolerance 10^{-5} , conservation ratio threshold 1.5, mirror-y relative-change threshold 0.5). Wilson 95% CIs are computed across the (SUT, seed) cells; raw outputs are committed (Table A1).

Per-mutant cross-family detection (R1). Four mutants are detected on every cell (30/30, CI [0.89, 1.00]) and five on none (0/30, CI [0.00, 0.11]), reproducing the C10 by-class localization: continuity $\rightarrow$ boundary/scale faults, symmetry $\rightarrow$ physical-channel/adjacency faults, node-permutation $\rightarrow$ none (these preserve node-relabeling invariance). One mutant is configuration-sensitive: mirror-y detects MA_permute_edges on the seed-replica family (20/20 over S0–S3) but not on the wider/deeper S4/S5 variants (0/10), a within-family rate of 20/30 (CI [0.49, 0.81]) invisible to a single checkpoint.

NS_double_scale severity sweep (R2). At every multiplicative scale $s \in \{1.1, 1.25, 1.5, 2.0, 4.0\}$ the detection rate stays at 0/6 (Wilson CI [0.00, 0.39]). A constant output factor crosses none of the three thresholds: the reference-relative conservation ratio stays below 1.5, mirror-y moves under 50% (source and mirror scale together), and the linear operation preserves node-permutation equivariance. The catalogue at the current thresholds is therefore structurally insensitive to multiplicative output scaling across the swept grid, not only at the canonical $2\times$ fault.

PC_zero_vy partial-fraction sweep (R3), a knife-edge symmetry blind spot. The refined grid resolves *where* detection collapses. At every partial fraction up to $p = 0.99$ the detection rate is 6/6 across the roster (Wilson CI [0.61, 1.00]); it falls to 0/6 (CI [0.00, 0.39]) only at the exactly-uniform $p = 1.0$. The transition is a step rather than a gradual decline; even 1%-asymmetric zeroing is caught on every checkpoint. The responsible detector is *node-permutation*: partial zeroing masks v_y on a node-index-selected subset, which is not relabeling-invariant; the symmetry MR never fires in this sweep (0/6 at all fractions). At $p = 1.0$ the fault becomes simultaneously permutation-invariant and mirror-y-symmetric (uniform $v_y = 0 = -0$ under reflection) and escapes every geometric detector at once. The canonical fault therefore sits in a measure-zero blind region that any partial severity exposes, a within-family finding showing a fault can be invisible to a geometric MR suite precisely when it shares the suite’s invariances.

Aggregate reading and Phase-3 unified catalogue. The 5-of-10 union detection rate from C10 reproduces across S0–S3 (4/10 on S4/S5). By-class localization (continuity $\rightarrow$ boundary/scale; symmetry $\rightarrow$ physical-channel/adjacency) replicates across the cylinder checkpoint roster (its cross-SUT transfer is examined below), and the insensitivity regions (multiplicative output scaling, the PC_zero_vy / adversarial-mutant blind subspace) are bounded with Wilson 95% CIs. This reflects the detectors’ coverage geometry: each MR scores a single invariant, so a fault is caught only when it perturbs a measured invariant and is structurally invisible when it preserves all of them; closing the blind region calls for additional MRs probing the uncovered directions, not more mutants of the same kind. Phase 3 additionally emits a 60-entry unified fault catalogue (Table A1): 10 executed canonical MGN mutants, 2 executed adversarial MGN mutants, 24 closed-form output-level PINN probes, and 24 closed-form output-level FNO probes. The precision/recall summary is node-permutation 1.00/1.00, conservation 1.00/0.81, and mirror-y 0.94/0.55; the same artifact reports Cliff’s δ and Wilcoxon signed-rank tests for the PINN MR-B ratios at $n = 15$ per PDE: Cliff’s $\delta = 0.73$ (large) and Wilcoxon $p = 0.0084$ for diffusion-vs-Burgers MR-B violation/floor ratios. This inter-PDE test is a physics-magnitude comparison (Dirichlet-zero Burgers BCs predict near-zero boundary residuals while Neumann-zero-flux diffusion BCs permit larger boundary residuals, so the magnitude difference is expected by construction) and is *not* a gate-reliability test. The valid reliability statistic is the per-PDE Wilson CI on the MR-B pass rate: Burgers 13/15 ([0.62, 0.96]) and diffusion 7/15 ([0.25, 0.70]). The PINN/FNO probes are not retrained mutant checkpoints.

Adversarial mutants (R4). Two adversarial mutants test whether the blind spot is a subspace, not a point. A1 is magnitude-driven, with node-permutation 0/6 and conservation 0/6; A2 escapes every detector. These results bound the detector set rather than establishing validated localization.

Cross-SUT keystone: the gate generates coverage. The principle’s sharpest prediction is that coverage tracks *admissibility*: transport the same MR suite to a second physics where the gate rules a relation inadmissible, and exactly that relation’s fault coverage should vanish while the admissible relations keep theirs. We test this on the primary-scale PhysicsNeMo airfoil (Section 5.5), with the same 10-mutant catalogue and predeclared thresholds, over five test trajectories (claim C36), and it holds. The non-zero angle of attack makes mirror-y domain-inadmissible, so the gate drops it, and with it vanishes exactly the coverage it alone supplied on the cylinder (the physical-

channel and adjacency faults). The admissible relations keep theirs: conservation still robustly detects the gross normalization fault (NS_skip_denorm, all five trajectories) and node-permutation still detects none; the one localization shared across both SUTs is continuity $\rightarrow$ normalization-scale faults, whose MR is admissible on both. The class-to-class mapping is SUT-specific not by accident but because the gate generates it from the SUT-specific admissible set. This is the duality in action: the same physics-based gate fixes both which verdicts are valid and which faults are detectable.

Falsifiable predictions, and what would refute them. The duality is stated to be refuted, not admired. It predicts that (i) a fault is detected iff it perturbs a measured invariant of an *admissible* MR, so coverage is computable from the admissible set before any fault is run; (ii) excluding an MR removes exactly its coverage and admitting one adds exactly its invariant’s; and (iii) blind spots are subspaces, closed only by MRs probing new invariant directions, never by more mutants of the same kind. Prediction (ii) is confirmed by the cross-SUT keystone above; (iii) by the knife-edge sweep, where detection collapses precisely at $p = 1.0$ as the fault becomes simultaneously permutation- and mirror-symmetric, and by the adversarial mutants. The principle would be refuted by coverage that survives an MR’s exclusion, a fault caught while perturbing no measured invariant, or a blind spot closed by same-kind mutants. The content is not the truism that a detector catches what it measures, but that a *physics-based validity predicate* (the gate that decides whether a relation’s verdict is meaningful) is what fixes the suite’s coverage: admissibility and detection power follow from one physical criterion, not two tuned separately. The evidence is qualitative (fault-class level) on two CFD SUTs; that boundary is the invited target, not a hedge.

Cross-program generalization. Reusing the committed Minimum-MR-SubSet kill matrices, the coverage-geometry reading reproduces across seven program types in three families: neural surrogates, classical numerical solvers (wave, point-kinetics, Burgers), and a production Monte-Carlo physics code (OpenMC). For each, fault coverage is the union of its admissible MR set’s per-class coverage; structural blind spots are present in six of the seven; and the class-to-class mapping is program-specific, extending the SUT-specificity finding from two CFD SUTs to seven program types. Detection rate tracks the admissible MR set rather than being uniform (minimal MR sets reach 10–20% with large structural blind spots, richer sets up to 100%), which is the coverage-generation structure the principle predicts. This is a read-only generalization check from committed evidence, not an end-to-end

execution of this pipeline on each program (claim C39).

End-to-end cross-program execution. Beyond that read-only reuse, this paper’s pipeline is executed end-to-end on five CPU-only SUTs imported read-only from the same sibling: four classical solvers (heat parabolic, wave hyperbolic, point-kinetics stiff-ODE, Burgers conservation law) and the sibling’s OpenMC subject run with the real OpenMC Monte-Carlo k -eigenvalue solver (multi-group infinite medium, verified against the closed-form k_∞), applying the four-condition gate, the typed verdict, and the coverage map to each program’s own domain MRs and seeded mutants. The gate is no rubber stamp: across the five it admits 38 candidate relations and rejects 7 (two on wave, three on point-kinetics, two on OpenMC) at the output-mapping condition, where a one-sided monotonicity relation is not falsified by a scrambled-observable probe, and defers none. Each admitted relation holds on the correct program (the in-domain, consistent region of the typed verdict) while the mutant runs populate the SUT-inconsistency region; every program keeps structural blind spots once the sibling’s declared-equivalent mutants are set aside, and the operator-class-to-coverage mapping is program-specific. Detection tracks the admissible set rather than being uniform (full-set mutation score 0.75 on heat, 0.20 on the other three solvers, and 0.56 on OpenMC, for the sibling’s committed mutant sets). The MRs and mutants are the sibling’s; only the gate and typed verdict are newly executed here, so this asserts no per-program reliability, no superiority, no new MR contribution, and, for OpenMC, no continuous-energy or whole-core reactor-physics result (claim C40).

5.8. Secondary baseline and external-scope audit

LLM baselines are secondary exploratory scope contrasts, not competitors. Without the rubric, three expert-LLMs proposed 25 candidates (24 unique); majority-vote rater models retained 4, downgraded 2, rejected 13, and deferred 6, for a 76% rejection/deferral rate and no novel retained MR family (an LLM-simulated rather than human-expert baseline). The generic-MR scope contrast admits only 3/13 domain-blind templates. The one-shot LLM baseline generated $K=8$ candidates; vendor-disjoint raters (glm-5.1, kimi-k2.6, deepseek-v4-flash) judged 7/8 panel-majority valid, with Fleiss kappa 0.077 (a small-sample paradox). Together these baselines show convergence on the same MR families while preserving the no-baseline-superiority boundary.

The Minimum-MR-SubSet audit records 70 real ABD instances, 20 real true-fault-class rows, and three SciML/PDE PASS_WITNESS rows. We also perform a **Minimum-MR-SubSet primary rerun**: the held-out cylinder-flow MGN witness returns PASS_WITNESS with $kstar = 6$, four active true fault classes, signature rank 2, and collapse false. This is real runtime evidence beyond read-only audit, without adding a second architecture/dataset.

5.9. Cross-family PINN extension ($K=6$ roster)

The cross-family executions test two claims the cylinder-flow study cannot test by itself: that the admissibility predicate is family-agnostic (the same four conditions, applied unchanged, produce typed decisions on pointwise PINNs and spectral FNOs), and that the conservation MR class yields full executable verdicts wherever its floor is calibratable: the FNO periodic conservation MR fails 24/24 against a calibrated case-level floor and the PINN reference-relative conservation passes 30/30, while the MGN open-boundary absolute variant is the one member of the class the gate correctly refuses. The deferral on cylinder flow is the fail-closed gate discriminating rather than a missing verdict.

To check whether the workflow is MeshGraphNets-specific, we use a $K=15$ PINN roster: fifteen Burgers and fifteen heat seeds. **MR-A** remains *vacuous by construction* for a pointwise MLP. The two non-trivial MRs do the cross-family work: **MR-B** passes on 13/15 Burgers seeds (violation/floor ratio mean 0.712, bootstrap CI [0.583, 0.875]; Wilson pass-rate CI [0.62, 0.96]) and is mixed on heat with 7/15 passing (mean 1.495, CI [1.142, 1.897]; Wilson pass-rate CI [0.25, 0.70]), two seeds near the residual-floor boundary fail on Burgers, while the Neumann-zero-flux heat BCs predict larger boundary residuals and yield a lower pass rate by construction; **MR-C** passes on all 15/15 of both PDEs (Wilson CI [0.80, 1.00]), with Burgers mean ratio 1.007 (bootstrap CI [0.997, 1.017]) and heat mean ratio 1.006 (CI [0.988, 1.022]). This is a two-PDE PINN seed roster rather than a PINN-vs-MGN benchmark or a generality claim across PINN architectures, PDEs, or training regimes.

5.10. FNO primary workflow upgrade ($K=6$)

The **FNO primary workflow upgrade** converts the earlier FNO roster into a second trained primary execution. Six small FNO-2D checkpoints cover Burgers and heat. For each checkpoint and four held-out generated periodic finite-difference cases, the workflow records rubric decisions,

source and follow-up tensors, mapped outputs, metric ledgers, and relation verdicts. Periodic integer translation is admitted and yields **24/24 translation passes** (maximum relative-L2 violation below 10^{-5}). The **periodic discrete-conservation MR** is admitted-with-reference-floor because the finite-difference target supplies a case-level channel-sum drift floor; the trained FNO outputs exceed that calibrated floor on **24/24 conservation failures**. The Dirichlet translation candidate is rejected for 6/6 SUTs because it changes the boundary-value problem and is not executed as an exact MR. This is **not only admissibility evidence**: it is a full rubric-to-verdict FNO execution with raw source/follow-up outputs and per-case ledgers, while remaining outside cylinder-flow evidence, performance benchmarking, reliability, and broad neural-operator generalization. Separately, the FNO admissibility roster is extended to $K = 15$ seeds per PDE so the periodic-translation admission decision is adequately powered: it is admitted on 15/15 per PDE (Wilson 95% CI [0.80, 1.00]), with near-machine-precision violation magnitudes (Burgers mean 4.5×10^{-8} , heat 1.4×10^{-7}).

5.11. Boundary of the evidence

The canonical blocked list is narrowed but still active: the evidence does not support external-dataset or geometry-independent pass/fail rates, Aero-GraphNet/DoMINO results, comparative superiority over baselines, general or real-world fault-detection rates, validated localization, runtime, reliability, or model accuracy claims.

6. Discussion

6.1. Interpretation of the scoped evidence

The value of the current study is not that every MR finds a new fault beyond rollout accuracy. Retained MRs provide relation-level evidence under explicit transformations; when a violation or deferral occurs, the MR card and verdict rule help distinguish model inconsistency, relation-domain boundary, numerical tolerance issue, and inconclusive evidence.

A natural objection is that the admissibility predicate merely re-describes the three initial outcomes. The added runs answer this: the symmetric-mesh run uses the predicate out-of-sample and fails an admitted relation; rollout accuracy shows the mirror-y violation answers a different question from

one-step error; and the primary empirical scope upgrade preserves these readings across a K=6 roster, three held-out trajectories, and larger denominators. The PINN roster, FNO primary workflow, and Minimum-MR-SubSet MGN/PINN reruns broaden applicability context while preserving the reliability and geometry-independent-rate boundary.

A second objection reads the deferred absolute-conservation relation as a gap in the end-to-end pipeline. The portfolio answers this directly: the conservation class produces full executable verdicts wherever its measurement floor is calibratable (FNO periodic conservation fails 24/24 against a calibrated floor; PINN reference-relative conservation passes 30/30), and the gate refuses execution precisely on the one variant where a verdict could not be separated from the operator’s own discretization error: MGN open-boundary absolute conservation.

The rubric averts two concrete misreadings: absolute discrete-divergence would have read as a conservation *pass* despite reference-field divergence, and asymmetric-mesh mirror-y would have read as a model fault rather than a geometric artifact.

6.2. *Boundary of claims*

The paper avoids four overclaims: that MRs always beat rollout accuracy; that NOETHER proves the physical validity of cylinder-flow MRs; that LLMs can automatically identify valid MRs; and that one MeshGraphNets family generalizes to all SciML surrogates. The safer claim is that a domain-validity-aware workflow makes MR identification and execution more auditable for a concrete class of SciML SUTs.

6.3. *Implications for SciML testing*

The scoped evidence shows how SciML testing can move from implicit expert checks to explicit MR assets that complement residuals, UQ, and accuracy by making transformations and verdict rules inspectable. The workflow targets a relation-indexed applicability map in relation space rather than residual space. We do not claim a completed applicability map in this paper; the present evidence is one bounded within-SUT point, and calibrated maps across SUTs, trajectories, and domain-violation scores remain future work.

7. Threats to Validity

We classify threats following Verdecchia et al. [32] and report against the Empirical Standards for software engineering research [33].

Construct validity. MR validity depends on the rubric, physical assumptions, BC compatibility, and tolerances; incorrect discrete operators or thresholds may create false violations/passes. The seeded-fault catalogue is author-implemented from the read-only witness taxonomy, so experimenter bias applies: detection rates do not bound real-world fault-detection effectiveness.

Internal validity. SUT setup, checkpoint differences, random seeds, mesh preprocessing, and runtime nondeterminism may affect verdicts. The experiment ledger records these details.

External validity. The evidence covers two CFD tasks on two official datasets: incompressible cylinder flow (four architectures: the MeshGraphNets K=6 roster, S4/S5 variants, PointMLP, and the PhysicsNeMo production implementation) and compressible airfoil flow, plus bounded cross-family PINN and FNO executions on Burgers and heat data. The airfoil result shows the predicate produces a physically correct, task-specific typed verdict structure rather than a fixed checklist, but it is a primary-scale K=6 roster (official architecture, 240 cells) with low surrogate accuracy at the local training budget (one-step relative L2 1.00); the Minimum-MR-SubSet reruns add a reproduced held-out cylinder-flow MGN witness and two trained PINN PDE witnesses as applicability checks, not general cross-family pass-rate estimates. Broader neural operators, mesh simulators, and PINN architectures remain outside the evidence. The cross-program breadth (seven program types reused read-only as committed kill matrices, with five SUTs, namely the four classical solvers plus a real multi-group OpenMC Monte-Carlo k -eigenvalue subject, additionally executed end-to-end through this paper’s gate and typed verdict on CPU) is a generalization check of the coverage-geometry reading on the sibling’s committed mutant sets: the MRs and mutants are the sibling’s domain assets, so it bounds neither per-program reliability nor a real-world detection rate, and makes no superiority or new-MR-contribution claim; the OpenMC subject is a multi-group infinite-medium criticality PUT, without continuous-energy whole-core scope.

Baseline fairness. Expert-MR, generic MR-generation, and LLM baselines may not be designed for SciML and use one prompt per generator. They should be interpreted as scope contrasts and candidate-generation comparators, not as defeated competitors.

Conclusion validity. Small samples, multiple MRs, and many transformation bins can produce unstable conclusions. The primary MGN scope upgrade expands the weakest denominators to K=6 x 3 trajectories x 10

mirror-y frames, $K=6 \times 3$ trajectories $\times$ 9 conservation transitions, and $K=6 \times 3$ exact-symmetry inputs. It removes the single-source-trajectory denominator within the stated scope.

Reproducibility. Some SUTs may depend on old runtimes or non-redistributable checkpoints. The paper should disclose such barriers and provide the most complete runnable package possible.

8. Conclusion

This paper presents domain-validity-gated MR identification as an auditable oracle-free testing workflow for SciML surrogates. The scoped pilots expand across $K=6$ MGN checkpoints and three held-out trajectories with 180 mirror-y, 162 conservation-transition, and 18 exact-symmetry cells; the verdict pattern replicates across same-task architectures, including wider/deeper MGN variants, a non-message-passing PointMLP network, and the NVIDIA PhysicsNeMo production implementation; and the readings are checked against bounded PINN/FNO executions, two primary trained-PINN witness reruns, an $O(h)$ operator-floor sweep, and severity/adversarial sweeps. The central claim is methodological: physically meaningful SciML MRs require explicit validity conditions, executable assets, raw evidence records, and relation-level verdicts.

CRedit authorship contribution statement

Meng Li: Conceptualization, Methodology, Software, Writing – original draft. **Xiaohua Yang:** Supervision, Formal analysis, Writing – review & editing. **Jie Liu:** Investigation, Validation. **Shiyu Yan:** Data curation, Visualization.

Declaration of competing interest

The authors declare no conflict of interest.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used generative AI assistants for code scaffolding, prose copy-editing, and reference cross-checking.

After using these tools the authors reviewed and edited the content as needed and take full responsibility for the content of the publication. No generative AI was used to fabricate, alter, or invent experimental data, results, or citations; every numerical claim is backed by a tracked artifact under `research_assets/runs/` and a claim-ledger entry.

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Data availability

The replication package contains the manuscript source, assets, ledger, and outputs, and is archived on Zenodo at <https://doi.org/10.5281/zenodo.20702952>. Minimum-MR-SubSet audit/rerun artifacts are under `research_assets/runs/` and cite commit `9ef862ec37335b4834d0a1fb38b4b613af702f34`. The MeshGraphNets checkpoints derive from the public DeepMind cylinder-flow benchmark; no proprietary data was used.

Appendix A. Claim-to-evidence map

Table A1: Compact claim-to-evidence map for the current manuscript.

Claim	Status	Evidence	Boundary
Domain-validity rubric	Method claim	Rubric asset and manuscript method section	Establishes an auditable screening rule; physical validity still depends on each relation's stated preconditions.
MR-card executable assets	Workflow claim	MR cards and validators	Some cards remain protocol assets pending matched SUT evidence.
Baseline admissibility contrast	Observed	Expert-MR, generic-MR, LLM-MR baselines and claim ledger	Quantifies an admissibility gap without ranking methods as competitors.
Minimum-MR-SubSet external-scope audit	Observed secondary audit	External-scope audit JSON	external witness evidence from 70 real rows and three SciML/PDE true-fault-class PASS_WITNESS rows; does not add new primary SUT executions to this paper.
Minimum-MR-SubSet primary rerun	Observed external runtime rerun	ABD witness rerun report	PASS_WITNESS, kstar = 6, four active true fault classes, max signature rank 2, collapse false; real held-out cylinder-flow MGN runtime, not a second architecture or dataset.
Minimum-MR-SubSet PINN primary reruns	Observed second-SUT/PDE reruns	ABD PINN witness rerun reports	trained Burgers2D and Diffusion2D PINN witnesses, PASS_WITNESS, kstar = 1 each, five active true fault classes each; one-seed witnesses, no cross-SUT rate.
Node-permutation sanity check	Observed pilot	Node-permutation metric ledger	One representation-contract path on one pilot case.
Conservation diagnostic	Diagnostic; absolute relation deferred	Conservation metric ledger and report	Reference-relative guard only; absolute conservation remains deferred.
Mirror-y OOD stress (PC6-mirror-y-ood-stress)	Observed bounded pilot	Mirror-y rate metric ledger and claim ledger	Failed on 10 of 10 recorded eval frames; not a reliability, accuracy, baseline, multi-SUT, exact-symmetry, or geometry-independent claim.
Primary empirical scope upgrade	Observed	Primary-scope report and per-trajectory/checkpoint manifests	K=6 x 3 trajectories x 10 mirror-y OOD-stress grid fails 180/180; K=6 x 3 trajectories x 9 conservation-transition grid passes 162/162; K=6 x 3 exact-symmetric-mesh input grid fails 18/18; not a single-source-trajectory estimate and not a cross-SUT rate.
LLM role boundary	Process boundary	Method and ethics sections	LLMs organize candidates; the rubric decides validity.
Rollout-accuracy diagnostic	Observed	Rollout-accuracy metric ledger	Same-SUT one-step median relative L2 0.0216; mirror-y is ~34x larger.
Exact mirror-y symmetric mesh	Observed	Symmetric-mesh metric ledger	Exact relation admissible and fails (rel L2 1.10) on a synthetic OOD mesh; binary equivariance evidence.
PC10-seeded-fault-detection	Observed	Seeded-fault metric ledger	Detector stress test: MRs catch 5/10 injected mutants with class-specific response patterns.
Multicheckpoint replication	Observed	E1 aggregate and per-SUT manifests in multicheckpoint/	K=6 checkpoints of one MeshGraphNets family; within-family replication.
Operator-floor resolution	Observed	Operator-floor sweep report	Log-log slope 0.984, 95% CI [0.975, 0.992] (interior 0.989), $R^2 = 0.9999$ over nine resolutions; calibrates the admissibility predicate on one mesh family.
Fault-detection robustness	Observed	E3 robustness report; Phase-3 unified catalogue	30-trial Wilson CIs per MGN mutant plus a 60-entry unified fault catalogue (10 canonical MGN + 2 adversarial MGN + 24 closed-form PINN output-level probes) + 24 closed-form FNO output-level probes), by-detector precision/recall with Wilson CIs, and Wilcoxon/Cliff effect-size tests; PINN/FNO probes are closed-form output-level probes.
S4/S5 variant primary workflow	Observed	Variant report and raw ledgers	Same-domain wider/deeper MGN variants: 2/2 node-permutation passes, 60/60 mirror OOD failures, 54/54 conservation-diagnostic passes, 6/6 exact-symmetric failures; not PhysicsNeMo/EchoWave or cross-dataset reliability evidence.

Claim	Status	Evidence	Boundary
PointMLP cylinder primary workflow	Observed	PointMLP report, checkpoint, and raw ledgers	Newly trained non-MGN row-wise coordinate network: 9/9 node-permutation passes, 10/10 mirror OOD failures, 9/9 conservation-diagnostic passes, 3/3 exact-symmetric failures, median rollout rel L2 0.0298; not PhysicsNeMo/EchoWave or production CFD evidence.
PhysicsNeMo MGN Object-A smoke workflow	Observed smoke-subset production-framework execution	PhysicsNeMo smoke report, newly trained checkpoint, raw outputs, and metric ledgers	NVIDIA PhysicsNeMo MeshGraphNet executes on a first-record official DeepMind cylinder_flow smoke subset: node permutation passes (relative L2 0.0), mirror-y is OOD stress, and conservation is reference-relative diagnostic only; not full production-scale PhysicsNeMo, AeroGraphNet, or DoMINO evidence.
FNO primary workflow upgrade	Observed	FNO workflow report, per-SUT ledgers, and raw source/follow-up outputs	Six trained FNO-2D checkpoints over Burgers/heat: 24/24 translation passes, 24/24 conservation failures under a periodic discrete-conservation MR, and 6/6 Dirichlet-boundary translation rejections; not only admissibility evidence.

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